place the equation in a shorter form, the two exponential terms can be combined:

$$X(\sigma,\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-(\sigma + j\omega)t}\, dt$$

Finally, the *location* in the complex plane can be represented by the complex variable, s, where $s = \sigma + j\omega$. This allows the equation to be reduced to an even more compact expression:

EQUATION 32-1
The Laplace transform. This equation defines how a time domain signal, $x(t)$, is related to an s-domain signal, $X(s)$. The s-domain variables, s, and $X(\ )$, are complex. While the time domain *may* be complex, it is usually real.

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

This is the final form of the Laplace transform, one of the most important equations in signal processing and electronics. Pay special attention to the term: e^{-st}, called a *complex exponential*. As shown by the above derivation, complex exponentials are a compact way of representing both sinusoids and exponentials in a single expression.

Although we have explained the Laplace transform as a two stage process (multiplication by an exponential curve followed by the Fourier transform), keep in mind that this is only a teaching aid, a way of breaking Eq. 32-1 into simpler components. The Laplace transform is a single equation relating $x(t)$ and $X(s)$, not a step-by-step procedure. Equation 32-1 describes how to calculate each *point* in the s-plane (identified by its values for σ and ω) based on the values of σ, ω, and the time domain signal, $x(t)$. Using the Fourier transform to *simultaneously* calculate all the points along a vertical line is merely a convenience, not a requirement. However, it is very important to remember that the values in the s-plane along the y-axis ($\sigma = 0$) are *exactly* equal to the Fourier transform. As explained later in this chapter, this is a key part of why the Laplace transform is useful.

To explore the nature of Eq. 32-1 further, let's look at several individual points in the s-domain and examine how the values at these locations are related to the time domain signal. To start, recall how individual points in the *frequency domain* are related to the time domain signal. Each point in the frequency domain, identified by a specific value of ω, corresponds to two sinusoids, $\cos(\omega t)$ and $\sin(\omega t)$. The real part is found by multiplying the time domain signal by the cosine wave, and then integrating from $-\infty$ to ∞. The imaginary part is found in the same way, except the sine wave is used. If we are dealing with the *complex* Fourier transform, the values at the corresponding negative frequency, $-\omega$, will be the complex conjugate (same real part, negative imaginary part) of the values at ω. The Laplace transform is just an extension of these same concepts.